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### SURGICAL STAPLER HAVING A POWERED HANDLE

#### Abstract

A powered handle for a surgical stapler can have a drive system including an electric motor. The powered handle can include a manual articulation mechanism to articulate a jaw assembly coupled to a reload shaft connected to the handle. The manual articulation mechanism can include a ball screw mechanism that translates an articulation member responsive to rotation of an articulation knob. The articulation mechanism includes a release function that allows the jaw assembly to return to a longitudinally centered orientation. The powered handle includes a battery pack serving as a power supply for the drive system. A control system can control actuation of the motor based on user inputs and operating parameters of the stapler. The powered handle can include a manual return mechanism.

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## Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 18/490,490, entitled “SURGICAL STAPLER HAVING A POWERED HANDLE,” filed Oct. 19, 2023 which is a continuation of U.S. patent application Ser. No. 17/591,356, entitled “SURGICAL STAPLER HAVING A POWERED HANDLE,” filed Feb. 2, 2022 which issued as U.S. Pat. No. 11,826,046, which is a continuation of U.S. patent application Ser. No. 16/821,830, entitled “SURGICAL STAPLER HAVING A POWERED HANDLE,” filed Mar. 17, 2020 which issued as U.S. Pat. No. 11,272,934, and which is a division of U.S. patent application Ser. No. 15/486,008, entitled “SURGICAL STAPLER HAVING A POWERED HANDLE,” filed Apr. 12, 2017 which issued as U.S. Pat. No. 10,610,225, and which claims the benefit of U.S. Provisional Patent Application Ser. No. 62/321,629, entitled “SURGICAL STAPLER HAVING A POWERED HANDLE,” filed Apr. 12, 2016. The above-referenced applications are incorporated by reference herein in their entireties.

## BACKGROUND OF THE INVENTION

### Field of the Invention

[0002] The present application relates generally to surgical occlusion instruments and, more particularly, to powered surgical staplers.

### Description of the Related Art

[0003] Surgical staplers are used to approximate or clamp tissue and to staple the clamped tissue together. As such, surgical staplers have mechanisms to ensure that tissue is properly positioned and captured and to drive staples through the tissue. As a result, this has produced, for example, multiple triggers and handles in conjunction with complex mechanisms to provide proper stapling of the clamped tissue. With these complex mechanisms, surgical staplers can have increased manufacturing burdens, as well as potential sources for device failure and confusion for the user. Thus, reliable stapling of clamped tissue without complex mechanisms is desired.

## SUMMARY OF THE INVENTION

[0004] In certain embodiments, a powered handle for a surgical stapling system is provided herein. The powered handle can comprise a drive system powered by a power supply to selectively actuate an actuation adapter. The powered handle can comprise a manual articulation mechanism to

selectively actuate an articulation adapter. The powered handle can further comprise a coupler having a bayonet coupling to simultaneously couple the articulation adapter and the actuation adapter to an articulation member and a drive member in a reload shaft.

[0005] In certain embodiments, the powered handle of the surgical stapling system comprises a control system to actuate the drive system responsive to user input from a movable trigger and a fire/return button on the powered handle. The control system can further vary an actuation profile of the drive system responsive to various operating parameters including the drive system operating torque, a longitudinal position of the actuation adapter, and identification of a jaw assembly length or configuration.

[0006] In certain embodiments, the powered handle of the surgical stapling system comprises a manual articulation system including a ball screw mechanism. The ball screw mechanism can allow continuous articulation of a jaw assembly of the stapling system within a predetermined articulation range. The ball screw mechanism can be biased to a longitudinally centered position and be rapidly centered through the use of a release mechanism.

[0007] In certain embodiments, a handle assembly for a surgical stapler is provided. The handle assembly comprises a handle body, an electric motor, an actuation shaft and a mechanical return mechanism. The handle body comprises a stationary handle and a trigger pivotably coupled to the handle body. The electric motor is disposed within the handle body. The actuation shaft is slidable within the handle body along a longitudinal axis and rotatable within the handle body about the longitudinal axis. The actuation shaft comprises a rack formed thereon. The actuation shaft is rotatable from a first position wherein the rack is operationally engaged with the electric motor to longitudinally slide the actuation shaft to a second position wherein the rack is disengaged from the electric motor and engaged with the manual return mechanism.

[0008] In certain embodiments, a handle assembly for a surgical stapler is provided. The handle assembly comprises a handle body, an electric motor, an actuation shaft, a motor gear, an auxiliary gear, a crown gear, a potentiometer, and a control system. The handle body comprising a stationary handle and a trigger pivotably coupled to the handle body. The electric motor is disposed within the handle body. The motor comprises an output shaft. The actuation shaft is slidable within the handle body along a longitudinal axis. The motor gear is coupled to the output shaft of the motor. The auxiliary gear is in driven engagement with the motor gear. The auxiliary gear is operatively engaged with the rack. The crown gear is mounted in the handle in meshed engagement with the motor gear. The potentiometer is coupled to the crown gear. The control system is electrically coupled to the trigger, the electric motor, and the potentiometer.

[0009] In certain embodiments, a handle assembly for a surgical stapler is provided. The handle assembly comprises a handle body, a power system, an actuation shaft, and an articulation mechanism. The handle body comprises a stationary handle and a trigger pivotably coupled to the handle body. The power system is within the handle body. The actuation shaft is operatively coupled to the power system. The actuation shaft is slidable within the handle body along a longitudinal axis. The articulation mechanism comprises a manually actuated articulation knob and an articulation adapter. The manually actuated articulation knob is positioned at a proximal end of the handle body and rotatable about the longitudinal axis. The articulation adapter is positioned at the distal end of the handle body. The articulation adapter is operatively coupled to the articulation knob such that rotation of the articulation knob about the longitudinal axis longitudinally slides the articulation adapter.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

[0010] FIG. 1 is a perspective view of an embodiment of surgical stapling system having an

embodiment of powered handle;

[0011] FIG. 2 is a side view of another embodiment of powered handle for the surgical stapling system of FIG. 1;

[0012] FIG. 2A is an exploded perspective view of the powered handle for the surgical stapling system of FIG. 2;

[0013] FIG. 3 is a partial cutaway perspective view of the powered handle of FIG. 2 with components removed to illustrate a drive system thereof;

[0014] FIG. 4 is a perspective view of an embodiment of drive system for the powered handle of FIG. 2;

[0015] FIG. 5 is a side view of an embodiment of drive system for the powered handle of FIG. 2;

[0016] FIG. 6 is a perspective view of an embodiment of power supply for the powered handle of FIG. 2;

[0017] FIG. 7 is a side view of the powered handle of FIG. 2 with the power supply in position for installation;

[0018] FIG. 8 is a perspective view of an embodiment of position indexing system for the powered handle of FIG. 2;

[0019] FIG. 9A is a perspective view of an embodiment of shaft coupler for the powered handle of FIG. 2 with an embodiment of shaft in a removed position;

[0020] FIG. 9B is a perspective view of an embodiment of shaft coupler for the powered handle of FIG. 2 with an embodiment of shaft in a partially inserted position;

[0021] FIG. 9C is a perspective view of an embodiment of shaft coupler for the powered handle of FIG. 2 with an embodiment of shaft in a fully inserted position;

[0022] FIG. 9D is a perspective view of an embodiment of shaft coupler for the powered handle of FIG. 2 with an embodiment of shaft in a retained position;

[0023] FIG. 10A is a cut-away side view of an embodiment of shaft coupler of FIG. 2 with an embodiment of shaft in a retained position;

[0024] FIG. 10B is a cut-away side view of an embodiment of shaft coupler of FIG. 2 with an embodiment of shaft in a latched position;

[0025] FIG. 11 is a cut-away side view of an embodiment of articulation mechanism of the powered handle of FIG. 2;

[0026] FIG. 12 is a perspective view of the articulation mechanism of FIG. 11 in an articulated position;

[0027] FIG. 13A is a cut-away side view of the articulation mechanism of FIG. 11;

[0028] FIG. 13B is a cut-away side view of the articulation mechanism of FIG. 11 with a release button depressed;

[0029] FIG. 13C is a cut-away side view of the articulation mechanism of FIG. 11 with the release button depressed and partially returned to a centered position;

[0030] FIG. 13D is a cut-away side view of the articulation mechanism of FIG. 11 with the release button depressed and returned to a centered position;

[0031] FIG. 13E is a cut-away side view of the articulation mechanism of FIG. 11 in the centered position with the release button partially released;

[0032] FIG. 13F is a cut-away side view of the articulation mechanism of FIG. 11 in the centered position;

[0033] FIG. 14 is a perspective view of an embodiment of manual return assembly for the powered handle of FIG. 2;

[0034] FIG. 15 is a perspective view of the manual return assembly of FIG. 14 with a decoupling mechanism actuated;

[0035] FIG. 16A is a side view of the manual return assembly of FIG. 14 with a return mechanism partially actuated;

[0036] FIG. 16B is a side view of the manual return assembly of FIG. 14 with the return

mechanism actuated through a full return stroke;

[0037] FIG. 17 is a perspective view of an embodiment of manual return assembly for the powered handle of FIG. 2;

[0038] FIG. 18 is a perspective view of the manual return assembly of FIG. 17 with a decoupling mechanism actuated;

[0039] FIG. 19A is a side view of the manual return assembly of FIG. 17 with a return mechanism partially actuated;

[0040] FIG. 19B is a side view of the manual return assembly of FIG. 17 with the return mechanism actuated through a full return stroke;

[0041] FIG. 20 is a schematic diagram of information and power flow for an embodiment of control system for the powered handle of FIG. 2; and

[0042] FIG. 21 is an operational sequence flow chart for an exemplary operational sequence of the powered handle of FIG. 2.

#### DETAILED DESCRIPTION OF THE INVENTION

[0043] With reference to FIGS. 1-2, an embodiment of surgical stapling system is illustrated. The illustrated embodiment of surgical stapler 10 comprises an elongate shaft 20, a jaw assembly 30, and a handle assembly 40. FIG. 1 illustrates the surgical stapler 10 with the jaw assembly 30 in an open configuration with an embodiment of powered handle having powered staple firing and powered jaw assembly articulation. FIG. 2 illustrates another embodiment of a powered handle 40 of the surgical stapler system 10 with the elongate shaft removed. The powered handle 40 of FIG. 2 has powered staple firing and manual jaw assembly articulation. In the illustrated embodiments, the shaft 20 and jaw assembly 30 can be freely rotated about a longitudinal axis defined by the shaft 20 by rotation of a rotation knob on the handle 40. In other embodiments, the stapling system can be configured to allow rotation of the jaw assembly about the longitudinal axis within a predefined range or a rotationally fixed jaw assembly.

[0044] With continued reference to FIG. 1, the illustrated embodiment of surgical stapler 10 can be sized and configured for use in laparoscopic surgical procedures. For example, the elongate shaft 20 and jaw assembly 30 can be sized and configured to be introduced into a surgical field through an access port or trocar cannula. In some embodiments, the elongate shaft 20 and jaw assembly 30 can be sized and configured to be inserted through a trocar cannula having a relatively small working channel diameter, such as, for example, less than 8 mm. In other embodiments, elongate shaft 20 and jaw assembly 30 can be sized and configured to be inserted through a trocar cannula having a larger working channel diameter, such as, for example, 10 mm, 11 mm, 12 mm, or 15 mm. In other embodiments, it is contemplated that certain aspects of the surgical staplers described herein can be incorporated into a surgical stapling device for use in open surgical procedures.

[0045] With continued reference to FIG. 1, as illustrated, the elongate shaft 20 comprises a generally tubular member. The elongate shaft 20 extends from a proximal end to a distal end. The elongate shaft 20 defines a central longitudinal axis, L, of the surgical stapler 10 extending between the proximal end 22 and the distal end 24.

[0046] With continued reference to FIG. 1, in the illustrated embodiment, the jaw assembly 30 is coupled to the elongate shaft 20 at the distal end of the elongate shaft 20. The jaw assembly 30 comprises a first jaw 32 and a second jaw 34 pivotally coupled to the first jaw 32. In the illustrated embodiment, the first jaw 32 is fixed to the distal end 24 of elongate shaft 20 such that it extends distally along the central longitudinal axis, L and is articulable with respect to the elongate shaft 20 responsive to an articulation mechanism in the handle 40. In an initial configuration, the first jaw 32 includes a plurality of staples 36 disposed therein within a reload 50. In other embodiments, the reload 50 can be integrated with the jaw assembly 30 such that the entire shaft assembly 20 and jaw assembly 30 with loaded staples define a single reload assembly. In some embodiments, staples can be initially positioned in the second jaw 34.

[0047] With continued reference to FIG. 1, in the illustrated embodiment, the jaw assembly 30 can

be actuated from an open configuration (FIG. 1) to a closed configuration to a stapling configuration by an drive member or beam that is longitudinally slideable within the elongate shaft. In an initial position, the beam can be positioned at the distal end **24** of the elongate shaft **20**. With the beam in the initial position, the second jaw **34** is pivoted away from the first jaw **32** such that the jaw assembly **30** is in the open configuration. The actuation beam engages the second jaw **34** upon translation of the actuation member or beam distally along the longitudinal axis L. Translation of the actuation beam distally from the initial position a first distance can actuate the jaw assembly from the open configuration to the closed configuration. With the jaw assembly **30** in the closed configuration, the actuation beam can be returned proximally the first distance to return the jaw assembly **30** to the open configuration. A distal end of the actuation beam can advance a staple slider configured to deploy staples from the first jaw **32** such that further translation of the actuation beam distally past the first distance deploys the plurality of staples **36** from the reload **50** in the first jaw **32**.

[0048] With continued reference to FIG. 1, in the illustrated embodiment, the handle assembly is coupled to the elongate shaft **20** at the proximal end of the elongate shaft **20**. As illustrated, the handle assembly **40** has a pistol grip configuration with a housing defining a stationary handle **42** and a movable handle **44** or trigger pivotably coupled to the stationary handle **42**. It is contemplated that in other embodiments, surgical stapler devices including aspects described herein can have handle assemblies with other configurations such as, for example, scissors-grip configurations, or in-line configurations. As further described in greater detail below, the handle assembly **40** houses a powered actuation mechanism configured to selectively advance an actuation shaft responsive to movement of the movable handle **44**.

[0049] In the illustrated embodiment, the surgical stapler **10** can include the plurality of staples **36** positioned in a disposable cartridge reload **50** while the jaw assembly **30** is configured to be reused with multiple staple cartridge reloads **50** in a single procedure. In the some embodiments, the elongate shaft **20** and jaw assembly **30** define a disposable reload shaft that is removably couplable to the handle assembly **40**. Accordingly, in the illustrated embodiment the handle assembly **40** includes a coupler **46** at the distal end thereof. The coupler **46** is adapted to engage the elongate shaft **20** of the surgical stapler **10**. The coupler **46** can have a bayonet connection having an outer connector that can removably couple to handle assembly **42** the elongate shaft **20**, and an inner connector that can removably couple the actuation shaft of the handle assembly **42** to the drive member of the elongate shaft **20**. Accordingly, the surgical stapler **10** can be configured such that the handle assembly **40** can be reused with multiple reload shafts **20** during a surgical procedure. It is contemplated that in other embodiments, the handle assembly and some portion of the elongate shaft can be reusable while a remainder of the elongate shaft in the jaw assembly define a disposable cartridge. In certain other embodiments, the handle assembly and the elongate shaft can be reusable while the jaw assembly defines a disposable cartridge. In still other embodiments, a jaw insert housing a plurality of staples can define a disposable cartridge while the remainder of the surgical stapler is reusable.

[0050] With reference to FIG. 2, an embodiment of powered handle for a surgical stapling system is illustrated. The powered handle can be used with various shaft reloads and cartridges such that the shaft configuration, jaw assembly configuration, and staple configuration can be selected for a particular procedure. The illustrated embodiment of handle provides powered (motor-driven) clamping and opening of the jaws and firing of the staple line. Articulation of the jaw assembly can be manually controlled by an articulation knob that the operator rotates. The motor is controlled by an embedded control system that dictates functionality of the handle during different stages of use.

[0051] With continued reference to FIG. 2, the powered handle **40** comprises a pistol-grip configuration with a stationary handle **42** and a movable handle **44** or trigger pivotably coupled thereto. A power supply **130** or battery can be positioned on a lower surface of the stationary handle. The powered handle **40** can further comprise a user control such as a fire or fire/reverse

button **150** to allow a user to selectively control a stapling sequence. The powered handle **40** can further comprise a redundant, manual return system **170** to allow a user to manually return the stapling system to an open configuration in the event of a powered system failure, control system failure, power supply failure, or “lockjaw” or other mechanical binding. The powered handle can further comprise a manual articulation mechanism including a rotatable articulation knob **190**. In the illustrated embodiment, the articulation knob **190** is positioned on the proximal end of the powered handle and is rotatable about an axis generally corresponding to the longitudinal axis of the stapling system.

[0052] With reference to FIG. 2A, the powered handle of FIG. 2 is illustrated in an exploded assembly view. Various elements of the illustrated embodiment of powered handle further discussed herein are identified in the exploded assembly view.

[0053] With reference to FIG. 3, a partial cut-away view of the powered handle is illustrated with a shaft **20** positioned in the coupler **46** of the handle. In the illustrated cut-away view, several components of the powered handle have been removed to clearly depict a drive system of the powered handle. In the illustrated embodiment, the drive system comprises a motor **112** positioned within the stationary handle **42**, a motor gear **114** positioned on an output shaft of the motor **112**, and an auxiliary gear **116** in driven engagement with the motor gear **114**. In some embodiments, the motor **112** is a brushed DC gearmotor. Advantageously, transmitting power through the auxiliary gear **116** can allow the motor **112** to be laterally centered within the stationary handle to enhance handle balance and user ergonomics. Furthermore, in some embodiments, the motor gear **114** and auxiliary gear **116** can be configured to provide a desired operational torque at the rack **122**. In some embodiments, the motor **112** can include a multigear transmission operationally coupled between the motor **112** and the motor gear **114** coupled to the auxiliary gear **116** to provide the desired operational torque. The motor **112** can be electrically coupled to the power supply **130** via a control system. The control system within the handle interfaces with the drive system to measure the position of the actuation shaft **120** and therefore the actuation of the jaw assembly.

[0054] The drive system is mounted to hardware that provides information to a control system including a microprocessor within the handle. This embedded system can control the speed and torque of the motor. It can also control functionality of the device based on user inputs (movement of the trigger and pressing of the FIRE/REVERSE button) and position of the drive system. The control system also can measure feedback from the motor to determine whether loads are too high to continue firing staples, or whether a reload cartridge lockout has been activated. It will also measure battery life and can limit the number of firings of the device.

[0055] With respect to FIG. 20, a schematic flow diagram indicating data and power flow for an exemplary control system for a powered handle is illustrated. In the illustrated flow diagram, the control system comprises the illustrated microcontroller. In various embodiments, the microcontroller can comprise an application specific integrated circuit or a general purpose microprocessor running application specific firmware and/or software. As illustrated, the microcontroller receives power and data regarding battery status from the batteries in the power supply. The microcontroller further receives data from various mechanical hardware of the stapler such as a motor driver and current monitor, an actuation rack position sensing mechanism, and a shaft connection and type monitor. The microcontroller further receives data from a user via a trigger position sensor, pushbutton switches, and a bluetooth communications transceiver. The control system can output a control signal to actuate the drive system of the powered handle through a motor driver. The control system can also output certain operational parameter information to a memory module on the power supply, and can output certain data for user viewing through LED lights on the handle and the bluetooth communications transceiver.

[0056] In certain embodiments, the control system is also configured to further define operational parameters of the powered handle. For example, by querying a memory module on the power supply or on the control system itself, the control system can detect whether the powered handle

has been used for more than a single procedure. In some embodiments, the stapling system is designed for use in a single procedure and is not designed for resterilization. Additionally, the control system can also query the memory modules on the power supply or the control system to detect a number of staple firings to assess whether sufficient battery power remains to complete an additional firing.

[0057] In certain embodiments, the control system is configured to detect tissue characteristics that can prevent staple firing. In some embodiments, the control system can monitor position, velocity, and supplied torque of the motor in the drive system. The control system can detect whether excessive torque is required to close the jaw assembly, if excess time is needed to close the jaw assembly, or if the jaws are closing at a low speed. These conditions may indicate that the tissue in the jaw assembly is too thick or too dense for the stapler to be effective. In certain embodiments, the control system can monitor the position of the actuation shaft with respect to time and evaluate this monitored position and time with respect to a baseline ‘zero load’ time reference position and time to assess the tissue characteristics such as thickness and density. In instances where the drive system exceeds predetermined operational parameters, the control system can indicate an error condition and stop a firing operation.

[0058] In some embodiments, the control system can provide user information over a bluetooth connection. The powered handle can include a low-power bluetooth transceiver to allow data regarding operational parameters such as battery status, number of remaining filings, and estimated tissue thickness to be displayed unobtrusively on a bluetooth-connected display.

[0059] With reference to FIG. 21, a schematic of an operational flow chart for an exemplary firing sequence of the control system is illustrated. As illustrated, the control system integrates user inputs from the trigger and firing button as well as hardware inputs from various sensors and monitors to advance the jaw assembly from a fully open condition to a fully closed condition to a firing sequence, then back to the fully open condition.

[0060] With reference to FIGS. 3-5, during powered operation, the auxiliary gear **116** is in meshed engagement with a rack **122** on an actuation shaft **120** extending longitudinally within the handle body. In the illustrated embodiment, the auxiliary gear is supported in a guide member through which the actuation shaft **120** slides. The guide member assists in maintaining meshed contact between the auxiliary gear and the rack **122**. A distal end of the actuation shaft **120** is freely rotatably coupled to an actuation adapter **124** that extends longitudinally into the coupler **46** at the distal end of the powered handle.

[0061] With the shaft **20** coupled to the coupler **46** of the powered handle **40**, the actuation adapter **124** connects to a drive member in the shaft **20** via a bayonet connection. Therefore, when the shaft **20** is attached to the handle **40**, the motor **112** and rack **122** will drive a drive member **22** coupled to the jaw assembly. Thus, the drive system within the handle comprises a “rack and pinion” design. Operation of the motor **112** responsive to a user's input will drive the actuation shaft **120** longitudinally forward and reverse to selectively actuate the stapler in closing, firing, or opening operations.

[0062] With reference to FIGS. 6 and 7, an embodiment of power supply **130** for the powered handle **40** is illustrated. The power supply **130** can be configured to deliver direct current to the powered handle motor and control system. In the illustrated embodiment, the stapler can operate at 12 V. The illustrated power supply can comprise four 3V lithium-ion batteries **132** connected in series to produce a 12V power supply. As illustrated, the batteries **132** are stacked in a 4 by 1 configuration in a plastic housing **134** to form the battery pack. In other embodiments, other numbers and configurations of individual battery cells can be used to form the battery pack. For example, in certain embodiments, the battery pack can be comprised of AA, AAA, or another standard or purpose-built single use or rechargeable chemistry battery. In the illustrated embodiment of powered handle **40**, the battery pack is located at the bottom of the stationary handle. Desirably, this positioning provides a stable surface to set the handle **40** on a flat surface. It

is contemplated that in other embodiments, the power supply can be positioned elsewhere in the handle, such as at a proximal end thereof (see, for example, the embodiment of FIG. 1).

[0063] With continued reference to FIGS. 6 and 7, in some embodiments, the power supply **130** can be packaged with the handle **40** but will not be installed before use. At the time of use, the user can install the battery pack by inserting it into a battery cavity **136** located at the bottom of the handle **40**. Advantageously, shipping the battery pack uninstalled can reduce an incidence of accidental battery discharge before use. Moreover, a removable battery pack can allow the stapler system be easily upgraded with a new battery as new battery technology becomes available. In other embodiments, the power supply can be packaged installed in the handle with a removable strip blocking electrical connection of the battery pack. In still other embodiments, the handle can be supplied with a power cable configured to be plugged into an AC or DC power source such as a wall socket, a USB connector, or another standard electrical connection.

[0064] In some embodiments, the power source further comprises a memory module such as a non-volatile memory that can store a digital record of the usage of the stapler. For example, the memory module can be configured to record details of each firing of the stapler including a periodic sampling of the battery voltage and motor current during firing, the sequence of states of the software state machine, any unexpected events that may have occurred, the shaft types that were used, the number of firings, the intervals between firings, and the model and serial number of the stapler handle. It can also record if the battery pack itself has been used so that users cannot reuse the battery pack.

[0065] In some embodiments, the powered handle **40** and associated power supply **130** can be configured for use in a single procedure and disposal following the procedure. The power supply **130** can include a power drain to reduce an opportunity for reuse. Following use in a surgical procedure, a user can remove the battery pack from the handle **40**. Removing the battery pack from the handle **40** can initiate draining the batteries. For example, after the battery pack has been used once a mechanical feature that can short circuit the battery by connecting the terminals to a low value resistor or an electrical feature can accomplish the same task with a circuit. Additionally, if the battery pack is left in the handle **40** after the surgical procedure is complete, in some embodiments, the control system of the handle is programmed to disable functionality and drain the battery pack after a maximum time limit.

[0066] With reference to FIGS. 2 and 8, an embodiment of position sensor mechanism for use in the powered handle is illustrated. In operation, rotation of the motor gear **114** correspondingly rotates a crown gear **142** mounted in the handle **40**. The crown gear **142** is coupled to a potentiometer such the position of the motor gear **114** and thus the actual position of the actuation rack can be determined based on the measuring changes in resistance at the potentiometer. In some embodiments, the potentiometer can be mounted on a circuit board **144** on which the control system can be positioned. While the illustrated embodiment includes a potentiometer-based position sensor mechanism, it is contemplated that in other embodiments, other position sensing mechanisms can be used, including, for example, use of a magnetic encoder with hall effect sensors, use of limit switches that activate when the actuation shaft has traveled a predetermined distance, use of optical systems such as photodiodes to measure travel of a pattern along the actuation shaft, or other position sensing systems.

[0067] With reference to FIG. 9A-9D, an operation sequence of engagement of a stapler shaft **20** with the coupler **46** of the handle is illustrated. In the illustrated embodiment, the reload shaft **20** to handle **40** connection comprises a bayonet style connection, in which a user axially aligns and inserts the reload shaft **20** into the handle **40** and rotates the reload shaft **20** approximately **90** degrees to connect. This bayonet connection operatively couples two mechanical functions of the reload shaft **20** to corresponding actuators of the handle **40**. When the bayonet connection is fully coupled, an articulation member within the shaft **20** is coupled to an articulation adapter of the handle and a drive member within the shaft **20** is coupled to the actuation adapter. Furthermore, the

handle **40** and shaft **20** can be configured with a latch mechanism at the coupler **46** to prevent a user from removing the shaft **20** once the actuation adapter and drive member has been activated. Moreover, the connection at the coupler **46** can include a reload identifying mechanism such that the control system of the handle can detect if a reload shaft is connected, and if so what the attached jaw length of the reload is. It is contemplated that the powered handle can be used with reload shafts **20** including different length jaw assemblies. For example, in some embodiments the same handle **40** can be used with either 45 mm or 60 mm length jaw assemblies. Thus, if the jaw assembly length is identified by the control system of the powered handle, the control system can direct a motor actuation profile for a firing stroke of the stapler corresponding to the identified length of the jaw assembly.

[0068] In FIG. **9A**, the shaft **20** is positioned in alignment with the coupler **46** on the handle, and a release knob of the coupler **46** is withdrawn to expose a bayonet channel **152** of the coupler **46** on a rotation insert of the coupler **46**. The shaft **20** can include a retention post **22** or boss positionable within the bayonet channel **152**. In the illustrated embodiment, the shaft includes two bosses positioned 180 degrees apart on the outer surface thereof and the coupler **46** includes a corresponding two bayonet channels **152**. It is contemplated that in other embodiments, other numbers and configurations of bosses and bayonet channels can be used to provide a desired connection strength and ease of alignment.

[0069] With reference to FIG. **9B**, the retention post **22** of the shaft is positioned within the bayonet channel **152**. With reference to FIG. **9C**, the reload shaft **20** has been rotated 90 degrees relative to the handle such that the retention post **22** of the shaft has reached a connected end of the bayonet channel **152**. With reference to FIG. **9D**, the release knob of the coupler is released to allow a retention recess **154** on the release knob to retain the retention post **22** of the reload shaft **20**.

[0070] With reference to FIGS. **10A-10B**, a cut-away side view of the coupler **46** with a reload shaft **20** is illustrated. The retention post **22** of the shaft is positioned within the retention recess of the bayonet channel. The actuation adapter **124** is coupled with a drive member **26** extending longitudinally within the shaft **20**. FIG. **10B** illustrates a lock-in or retention mechanism that operates upon initial distal advancement of the actuation adapter **124**. As illustrated, a locking member **24** is pivotably coupled to a proximal end of the shaft **20**.

[0071] With continued reference to FIG. **10B**, the locking member **24** can include a ramped or tapered lock surface at a proximal edge thereof. As illustrated in FIG. **10A**, the shaft **20** is in a coupled, but unlocked configuration with respect to the coupler **46**. In the coupled, unlocked configuration, the shaft **20** can be removed from the coupler **46** through the bayonet connection by a reverse of the sequence of operations of FIGS. **9A-9D**. Once the actuation adapter **124** is advancing to operate the stapler, the actuation adapter **124** interacts with the ramped surface of the locking member **24** to advance the locking member radially outward into a locked position. In the locked position (FIG. **10B**), the locking member **24** engages a locking ledge on the coupler **46** to lock in the shaft. With the shaft **20** locked in with respect to the handle **40**, the shaft **20** cannot be removed from the handle **40** until the actuation adapter **124** has been returned to a fully proximally retracted position (typically corresponding to a return to a jaws open configuration following a full closure and stapling cycle of the jaw assembly).

[0072] Thus, the “lock In” feature prevents a user from removing the shaft from the handle once the drive member **26** has been driven forward. Once the locking member **24** is situated in the slot or ledge of a rotation insert of the coupler **46**, a release knob of the coupler **46** is unable to be pulled back. This locking action on the coupler prevents the user from rotating the shaft **20** out of the bayonet connection of the coupler **46** once actuation of the stapler has begun.

[0073] With reference to FIGS. **11**, **12**, and **13A-13F**, an embodiment of articulation mechanism for the powered handle **40** is illustrated. In the illustrated embodiment, the handle can articulate the jaw assembly at the distal end of the shaft up to 45° in a fully articulated position in either direction relative to a longitudinally centered position. In some embodiments, the powered handle uses a

manual articulation mechanism including a series of components coupled to the manually actuated articulation knob **190** at the proximal end of the handle. In other embodiments, the manually actuated articulation knob and certain associated elements of the articulation mechanism can be positioned in other locations on the handle such as adjacent a distal end of the handle.

[0074] With reference to FIGS. **11** and **12**, the articulation mechanism is coupled to an articulation member **206** extending longitudinally within the reload shaft when the reload shaft is coupled to the handle. Actuation of the articulation mechanism longitudinally translates the articulation member **206** proximally or distally relative to the shaft to articulate the jaw assembly at the distal end of the shaft.

[0075] With reference to FIG. **11**, the articulation mechanism comprises a ball screw **192** having at least one helical groove or thread **195** in which one or more ball bearing **194** can ride. In the illustrated embodiment, the articulation mechanism comprises two ball bearings **194** that are engageable in two threads **195**. The ball bearings **194** are positioned in ball bearing apertures **189** in a ball sleeve **191** positioned radially outwardly of the ball screw **192**. The ball bearings **194** are maintained in the threads **195** by a release sleeve **196** positioned radially outward of the ball bearings **194**. Rotation of the articulation knob **190**, which is coupled to the ball sleeve **191** such as by connecting pins **193**, rotates the ball sleeve **191** about an axis of rotation, causing the ball bearings **194** to travel within the threads **195** and correspondingly longitudinally translate the ball screw **192**. Articulation of the jaw assembly is accomplished by rotating the articulation knob **190** to correspondingly rotate the ball sleeve **191** and the ball bearings **194** about the axis of rotation while their longitudinal position is fixed along the axis of rotation. The ball bearings **194**, which are engaged in the threads **195** of the ball screw **192** will then translate the ball screw **192** forward and reverse along the axis of rotation. In the illustrated embodiment, the ball sleeve **191** is generally tubular, having a cavity formed therein, and a portion of the ball screw **192** is positioned within the cavity and translates longitudinally within the cavity. While the illustrated embodiment of articulation mechanism includes two ball bearings engageable threads in a ball screw, it is contemplated that in other embodiments, the articulation mechanism can have fewer or more than two ball bearings such as, for example, a single ball bearing positioned in a single helical screw or three or more ball bearings in a corresponding number of helical threads.

[0076] With reference to FIGS. **11** and **12**, the ball screw **192** extends to a distal end **200** coupled to a pair of articulation links **202**. The articulation links **202** are spaced apart from one another, which desirably allows them to be positioned radially outwardly of the drive system and actuation shaft within the handle. As illustrated in FIG. **12**, the articulation links **202** can comprise a mating feature such as a slot formed therein to allow them to be keyed into a corresponding mating feature such as a post extending radially inwardly from the handle body. The slots can stabilize the articulation links relative to the handle and interaction of the handle posts with ends of the slots can define a range of articulation for the articulation mechanism. The distal ends of the articulation links **202** can be rotatably coupled to the articulation adapter **204**, which can be positioned coaxially radially outwardly of the actuation adapter at the distal end of the handle. This rotational coupling can include an articulation bearing **205** having relatively low friction properties. This articulation bearing **205** can facilitate rotation of a coupled reload shaft relative to the handle assembly and longitudinal movement of the articulation adapter **204** during operation of the articulation mechanism. While the illustrated embodiment of articulation mechanism includes two articulation links laterally offset from the actuation mechanism within the handle, it is contemplated that in other embodiments, the articulation mechanism can have fewer or more than two articulation links such as, for example, an articulation link or three or more articulation links.

[0077] With continued reference to FIG. **12**, the articulation adapter **204** can be connected to the articulation member **206** in the shaft by a bayonet connection when the shaft is coupled to the handle. The threads **195** can be configured such that moving the ball screw proximally will articulate the jaw assembly to the left when viewed from the handle relative to a longitudinally

centered position and moving the ball screw **192** distally will articulate the jaw assembly to the right when viewed from the handle relative to the centered position.

[0078] Advantageously, since the helical threads **195** of the ball screw **192** are continuous, the articulation mechanism can allow the jaw assembly to be articulated to virtually infinite angular positions between a desired operational range. In some embodiments, the articulation mechanism can be configured to provide an articulation operational range from  $-45^{\circ}$  to  $+45^{\circ}$  of the jaw assembly relative to a longitudinally centered position defined by the longitudinal axis of the shaft. In other embodiments, the articulation mechanism can be configured to provide other operative articulation ranges including ranges providing more than  $\pm 45^{\circ}$  of articulation or those providing less than  $\pm 45^{\circ}$  of articulation. In some embodiments, the articulation mechanism can be configured to provide articulation in a single direction relative to a longitudinally centered position.

[0079] In some embodiments, the pitch of the threads **195** on the ball screw **192** is variable. For example, the threads **195** can include a relatively low pitch towards an end of the threads to advantageously provide a larger mechanical advantage when the jaw assembly can require more force to articulate. The threads **195** can include a relatively higher pitch towards a center of the threads to allow rapid movement with a relatively lower mechanical advantage where the jaw assembly can require a lower force to articulate. In other embodiments, the threads **195** include a constant pitch such that rotation of the articulation knob results in a proportional amount of articulation of a jaw assembly of the stapler that does not vary over the articulation range of the articulation mechanism. Desirably, such a constant pitch thread ball screw can result in an easily predictable response during operation of the actuation mechanism.

[0080] With reference to FIGS. **13A-13F**, the articulation mechanism can comprise a release mechanism that allows the articulation mechanism to advantageously be reset to the longitudinally centered position from any articulated position. The release mechanism is operated by user pressing a release button **198**. In the illustrated embodiment, the release button **198** is positioned radially nested within the articulation knob **190**.

[0081] With reference to FIG. **13B**, operation of the release button **198** will distally advance the release sleeve **196**. A radially inner surface of the release sleeve **196** is stepped to include an engagement surface **186** having a relatively small inner diameter and a release surface **188** having a relatively larger inner diameter with a smooth ramp between the engagement surface and the release surface. In operation, the engagement surface of the release sleeve maintains the ball bearings **194** in the threads **195** of the ball screw **192**. Once the release button **198** is pushed, the engagement surface is distally advanced, allowing the ball bearings **194** to disengage from the threads **195** and advance radially outward through the ball bearing apertures **189** in the ball sleeve against the release surface.

[0082] With reference to FIGS. **13C** and **13D**, with the ball bearings **194** disengaged from the threads **195**, the articulation mechanism can be biased to a centered position. In some embodiments, the ball screw **192** is biased to a centered position by a biasing member such as two springs **197** and spring force from the shaft. The ball bearings **194** positioned in the centered position (FIG. **13D**) along the threads **195** corresponds to a longitudinally centered position of the jaw assembly.

[0083] With reference to FIGS. **13E-13F**, once the release button **198** is allowed to return to an undisturbed configuration, release sleeve **196** is retracted proximally (indicated by arrows **199**) by a spring. Proximal movement of the release spring **196** forces the ball bearings **194** into engagement with the threads **195** of the ball screw. Thus, the articulation mechanism can then be used to articulate the jaw assembly from the longitudinally centered position, or the stapler can be used with the jaw assembly in the longitudinally centered position.

[0084] With reference to FIGS. **14**, **15**, and **16A-16B**, an embodiment of manual return mechanism for the powered handle is illustrated. A manual return mechanism can advantageously provide a redundant return mechanism in the event of a power supply failure, other powered component

failure, or mechanical failure or binding.

[0085] With reference to FIGS. **14** and **15**, the manual return mechanism includes two separate, independently operable subassemblies that are operated in sequence to return the actuation shaft **120** to a proximal-most position within the handle, which corresponds to the open configuration of the jaw assembly. As illustrated, the manual return mechanism **170** comprises a shaft rotation mechanism and a shaft retraction mechanism. In operation, when it is desirable to manually return the stapler to the open configuration, the shaft rotation mechanism is initially operated.

[0086] With reference to FIGS. **14** and **15**, to operate the shaft rotation mechanism of the manual return mechanism **170**, a user pulls a disengagement tab **172** positioned on an outer surface of the handle. The disengagement tab **172** has a disengagement rack **174** formed thereon. The disengagement rack **174** is in meshed engagement with a shaft rotation rack **176** formed on a shaft rotation collar **176**. The actuation shaft **120** extends through the shaft rotation collar **176** and is slideable therethrough. Thus pulling the disengagement tab **172** rotates the actuation shaft **120** approximately **90** degrees about the longitudinal axis thereof. This rotation positions the rack **122** of the actuation shaft out of engagement with the auxiliary gear **116** of the drive system. Moreover, in some embodiments, removal of the disengagement tab **172** from the handle can also disengage the power supply from the drive system or otherwise disengage the control system to prevent further powered operation of the powered handle. Additionally, the shaft rotation mechanism can be configured to be operated a single time only. For example, in the illustrated embodiment a return pawl **182** on the shaft retraction mechanism can comprise an interference lobe **183** sized and configured to interfere with the drive system to prevent closure of the return lever and rotation of the actuation shaft **120** back into engagement with the auxiliary gear once the shaft rotation disengagement tab **172** has been pulled. Thus, once the shaft rotation mechanism has been operated, the handle can be disabled from further use.

[0087] With reference to FIGS. **15** and **16A-16B**, once the shaft rotation mechanism has been operated, the shaft retraction mechanism can be operated to return the actuation shaft proximally within the handle. Removal of the disengagement tab **172** from the handle exposes a return lever **180** on the powered handle. The return lever **180** is pivotably coupled to a return pawl **182** at a pivot joint **184**. When the rack **122** of the actuation shaft **120** was rotated out of engagement with the drive system, it was rotated into engagement with the shaft retraction mechanism. The return lever **180** can be rotated through one or a series of return cycles (FIGS. **16A**, **16B**) to engage the return pawl **182** with the rack **122** on the actuation shaft **120** and retract the actuation shaft **120** proximally within the handle in a ratchet-type operation.

[0088] With reference to FIGS. **17**, **18**, and **19A-19B**, another embodiment of manual return mechanism for the powered handle is illustrated. The components and operation of the manual return mechanism **170'** are similar to that described above with respect to the manual return mechanism **170** of FIGS. **14**, **15**, and **16A-16B**. However, in use of the manual return mechanism **170'**, removal of a disengagement tab **172'** from the handle assembly exposes a shaft rotation collar **176'** having a rotation lever **178'** protruding therefrom. With the handle assembly in powered operation, the disengagement tab **172'** covers the shaft rotation collar **176'** on an outer surface of the handle. Once the disengagement tab has been removed, a user can then manipulate the rotation lever **178'** to rotate the actuation shaft **120** such that the shaft retraction mechanism can be operated to return the actuation shaft proximally within the handle. The shaft retraction mechanism of the manual return mechanism **170'** includes the same ratchet-type operation as that discussed above with respect to the manual return mechanism **170**. Desirably, in some handle configurations, the rotation lever **178'** can provide enhanced mechanical advantage to facilitate rotation of the actuation shaft as compared to the shaft rotation mechanism including a disengagement rack **174** of FIGS. **15**, **15**, and **16A-16B**.

[0089] Although this application discloses certain preferred embodiments and examples, it will be understood by those skilled in the art that the present inventions extend beyond the specifically

disclosed embodiments to other alternative embodiments and/or uses of the invention and obvious modifications and equivalents thereof. Further, the various features of these inventions can be used alone, or in combination with other features of these inventions other than as expressly described above. Thus, it is intended that the scope of the present inventions herein disclosed should not be limited by the particular disclosed embodiments described above, but should be determined only by a fair reading of the claims.

## Claims

1. A handle assembly for a surgical stapler, the handle assembly comprising: a handle body comprising a stationary handle and a trigger pivotably coupled to the handle body; an electric motor disposed within the handle body; an actuation shaft slidable within the handle body along a longitudinal axis and rotatable within the handle body about the longitudinal axis, the actuation shaft comprising a rack formed thereon; and a manual return mechanism comprising a disengagement tab positioned on an outer surface of the handle body; wherein the actuation shaft is rotatable by removal of the disengagement tab from the handle body from a first position wherein the rack is operationally engaged with the electric motor to longitudinally slide the actuation shaft to a second position wherein the rack is disengaged from the electric motor.
2. The handle assembly of claim 1, wherein the disengagement tab comprises a disengagement rack formed thereon, and wherein the actuation shaft comprises a shaft rotation collar disposed thereon, the shaft rotation collar comprising a shaft rotation rack in engagement with the disengagement rack.
3. The handle assembly of claim 1, wherein the manual return mechanism further comprises: a shaft retraction mechanism configured to proximally retract the actuation shaft upon removal of the disengagement tab from the handle body.
4. The handle assembly of claim 3, wherein the shaft retraction mechanism comprises: a return lever; a return pawl; and a pivot joint pivotably coupling the return lever to the return pawl.
5. The handle assembly of claim 4, wherein the return pawl is engaged with the rack with the actuation shaft in the second position.
6. The handle assembly of claim 5, wherein actuation of the return lever advances the actuation shaft proximally with the actuation shaft in the second position.
7. The handle assembly of claim 3, wherein the shaft retraction mechanism is configured to prevent rotation of the actuation shaft from the second position to the first position.
8. A handle assembly for a surgical stapler, the handle assembly comprising: a handle body comprising a stationary handle and a trigger pivotably coupled to the handle body; an electric motor disposed within the handle body; an actuation shaft slidable within the handle body along a longitudinal axis and rotatable within the handle body about the longitudinal axis, the actuation shaft comprising a rack formed thereon; and a manual return mechanism comprising a disengagement tab on the handle body and a shaft rotation collar having a rotation lever covered by the disengagement tab; wherein the actuation shaft is rotatable by the rotation lever from a first position wherein the rack is operationally engaged with the electric motor to longitudinally slide the actuation shaft to a second position wherein the rack is disengaged from the electric motor.
9. The handle assembly of claim 8, wherein the manual return mechanism further comprises: a shaft retraction mechanism configured to proximally retract the actuation shaft upon rotation of the actuation shaft to the second position.
10. The handle assembly of claim 9, wherein the shaft retraction mechanism comprises: a return lever; a return pawl; and a pivot joint pivotably coupling the return lever to the return pawl.
11. The handle assembly of claim 10, wherein the return pawl is engaged with the rack with the actuation shaft in the second position.
12. The handle assembly of claim 11, wherein actuation of the return lever advances the actuation

shaft proximally with the actuation shaft in the second position.

**13.** The handle assembly of claim 9, wherein the shaft retraction mechanism is configured to prevent rotation of the actuation shaft from the second position to the first position.

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